

Will September CPI rise more than 0.2%?

Expected Parrot

Open-source research tools

Vorhersage | Forecast research report

2026-09-16T22:31:28.278519Z

Assessment

Our sealed estimate is 66.9%; Kalshi's opening midpoint was 98.5%. Current gasoline data changed the model sharply, but a 31.6-point disagreement remains. This non-weather case exposes a concrete process weakness: we adjusted the expected CPI increase without validating the uncertainty of the complete model.

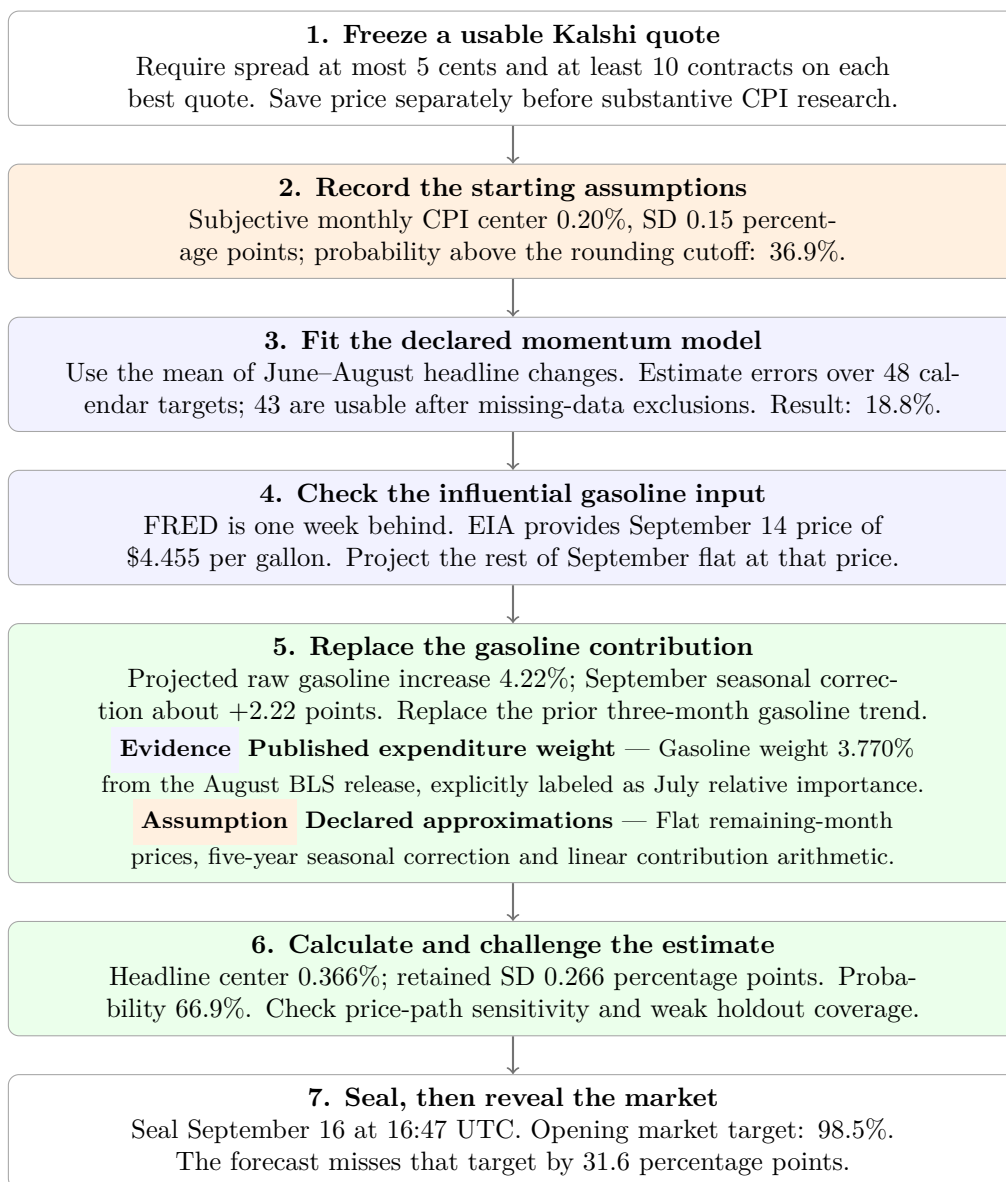
What counts as Yes?

Kalshi contract KXCPI-26SEP-T0.2 concerns the signed, seasonally adjusted monthly increase in the all-items Consumer Price Index for All Urban Consumers (CPI-U) for September 2026. It is not annual inflation or core inflation. The BLS figure must be greater than 0.2% when published to one decimal place: ordinarily 0.3% or higher.

The release is scheduled for October 14 at 8:30 a.m. Eastern. The market-specific rules close trading five minutes earlier. Delay, shutdown, no-data fallback and review provisions are retained in the contract record. Our numerical model assumes an ordinary release; it does not separately model those exceptional cases.

Methodology at a glance

The market quote stayed in a separate evaluator store until the forecast was sealed. The reference window, formula and gasoline adjustment were specified before data extraction. Blue boxes contain evidence, amber boxes assumptions, and green boxes calculated results.



Two model stages within one research pass. The stages are not independent experiments; gasoline information had already been collected when the intermediate momentum result was journaled.

The recorded estimates

Our initial estimate was a subjective starting model. The momentum stage replaces it with a reproducible historical calculation. The gasoline stage incorporates current information about an influential component. All three estimates were recorded before the market quote was revealed.

Stage	Probability	Gap below opening midpoint
Subjective initial model	36.9%	61.6 percentage points
Historical headline momentum	18.8%	79.7 percentage points
After current gasoline adjustment	66.9%	31.6 percentage points
Opening Kalshi quote	98–99%; midpoint 98.5%	Frozen development target

Selection, chronology and liquidity

The contract was chosen without viewing its price. Its opening quote was captured at 16:41:57 UTC on September 16. The initial estimate and plan followed at 16:42. The two calculated checkpoints and seal were saved at 16:47. Reveal occurred at 22:28:05 UTC; the later quote was still 98–99%. The research record does not use intervening information to rewrite the estimate.

The opening best bid had 595.01 contracts and the best ask 2,954.5. There were 5,140.01 bid-side contracts within two cents of the best bid. Both snapshots passed the original ten-contract screen, with unchanged contract terms. This is materially stronger quote depth than the weather examples, although two snapshots do not establish sustained liquidity or market calibration.

This was the same assistant context, with broad macroeconomic information from an earlier Fed case already present. The evaluator store was not inspected before sealing, but no independent context or enforced access barrier was used. Contract thresholds were visible. Research and monetary costs were not separately metered.

Why the momentum-only estimate was low

The fixed baseline averages the preceding three monthly headline CPI changes. Calculated from the downloaded BLS indices, June was -0.4225% , July $+0.0737\%$ and August $+0.3960\%$. Their mean is just $+0.0157\%$. A large June decline therefore continues to pull down this simple September extrapolation.

We compared the same three-month rule with subsequent monthly changes for every target month from September 2022 through August 2026. October 2025 has no published headline index in the downloaded series. That breaks both the target change and several lagged changes, leaving five excluded target months, October 2025 through February 2026. No values were interpolated.

The 43 remaining errors have mean -0.0010 percentage points and sample standard deviation 0.2625. Adding the estimated mean error and its simple sampling-uncertainty allowance gives a baseline center of 0.0147% and spread of 0.2655 percentage points. Under a normal approximation, that implies an 18.8% chance above the effective 0.25% rounding cutoff.

Source: BLS CPI indices via Federal Reserve Bank of St Louis

Source: BLS August2026 CPI release

Source: BLS seasonal adjustment FAQ

What the gasoline research changed

The FRED gasoline download ended with September 7. Checking EIA's original spreadsheet found the September 14 all-grades price of \$4.455 per gallon, compared with \$4.295 a week earlier. The primary-source check prevented us from treating a lagging mirror as current.

We carried each weekly price forward to the next observation and the latest price through September 30. The resulting calendar-day averages are \$4.1912 for August and \$4.3681 for projected September, a 4.22% increase. This is an explicit projection of unobserved days, not a completed monthly observation.

Gasoline normally has a seasonal pattern. Across all five Septembers from 2021 through 2025, the median difference between seasonally adjusted and raw gasoline changes was +2.22 percentage points. Applying that proxy yields about 6.44% seasonally adjusted gasoline growth. The preceding three-month gasoline trend was -2.88%.

Replacing that old gasoline contribution with the new one, using the published 3.770% expenditure weight, adds 0.3515 percentage points to the headline estimate. The August release labels this weight as July relative importance; we preserve that lag explicitly. Rolling it forward approximately gives a 68.0% sensitivity result rather than the selected 66.9%. The linear contribution calculation is an approximation to BLS aggregation.

Input or calculation	Value
August daily-step gasoline average	\$4.1912/gallon
Projected September average	\$4.3681/gallon
Projected raw gasoline change	+4.22%
Median September seasonal correction	+2.22 percentage points
Projected adjusted gasoline change	+6.44%
Prior three-month adjusted gasoline trend	-2.88%
Replacement effect on headline CPI	+0.3515 percentage points

Source: EIA weekly US all-grades all-formulations retail gasoline price history

Source: FRED EIA weekly gasoline mirror

Source: EIA retail gasoline survey methodology

Source: BLS CPI indices via Federal Reserve Bank of St Louis

Source: BLS CPI August2026 Table1

Source: BLS seasonal adjustment FAQ

From a CPI point estimate to a probability

The final headline center is $0.0147\% + 0.3515\% = 0.3662\%$. We retain the baseline model's 0.2655-percentage-point spread, as declared before retrieval. Assuming normal forecast errors, the chance of an unrounded change at least 0.25% is 66.9%. The empirical historical-error version gives 72.1%. Neither figure is validated calibration for this gasoline-adjusted model.

Keeping an old spread is a modeling choice, not a mathematical requirement. It avoids automatically claiming that additional research increases precision, but it also creates a mismatch: the center uses current gasoline information while the error distribution comes from a model that ignored it. That mismatch is a central limitation of this exercise.

Source: BLS CPI indices via Federal Reserve Bank of St Louis

Source: EIA weekly US all-grades all-formulations retail gasoline price history

Source: BLS CPI August2026 Table1

Source: Kalshi CPI contract terms

Source: BLS seasonal adjustment FAQ

Sensitivity and a failed calibration check

A fixed chronological diagnostic used the first 36 calendar months for fitting and the next 12 for checking. Missing data left only seven usable holdout forecasts. Their RMSE was 0.502 percentage points, and a nominal 90% interval covered just four of seven outcomes. This is a warning about instability, not evidence that the primary normal approximation is reliable.

Before revealing the market, we also computed a wider-distribution diagnostic using that holdout RMSE as a stand-in for spread. It gives 59.2%. Seven errors cannot establish a calibrated alternative distribution; we report this to show how much the uncertainty choice matters.

If prices for the unobserved remainder of September are 5% below or above the flat-price assumption, the primary model gives 53.4–78.5%. Using each of the five individual historical seasonal corrections gives 65.4–68.1%. Shifting the final headline center by ± 0.05 percentage points gives 59.8–73.4%. These are assumption sensitivities, not confidence intervals.

Source: BLS CPI indices via Federal Reserve Bank of St Louis

Source: BLS seasonal adjustment FAQ

Source: EIA weekly US all-grades all-formulations retail gasoline price history

Source: BLS CPI August2026 Table1

Source: Kalshi CPI contract terms

A substantial miss against the chosen development target

Kalshi's opening midpoint was 98.5%. Our final 66.9% estimate is 31.6 percentage points lower and 31.1 points below the opening bid. The later midpoint was unchanged. With market agreement as the development criterion, this is a substantial miss even though the gasoline research moved the model closer.

The outcome remains unresolved. Closer agreement is not a resolution score, and these sequential calculations do not identify the causal value of research. We have no basis to claim the market is wrong. Equally, the market quote alone does not tell us which of our inputs or distributional assumptions is responsible for the difference.

A post-reveal Gaussian diagnostic makes the scale explicit. Holding our spread fixed would require a center near 0.826% to reach 98.5%. Holding our center fixed would require a spread near 0.054 percentage points. Those are arithmetic descriptions of disagreement, not recommended parameters or new forecasts.

What should change on the next fresh question

Build forecasts for the major CPI components, including shelter/core, food and energy, with dated inputs available at the intended forecast time. Estimate the errors of that complete model using original-release outcomes. A model with a new center and an unrelated old error distribution is not enough.

Make freshness checks explicit: compare mirrored datasets with the primary release and record missing periods. Use the published current seasonal factors where available instead of treating a five-year median as an exact factor. Model the remaining-month gasoline path as a source of uncertainty.

This run uses revised historical CPI indices, a coarse seasonal proxy and a small, weak holdout. It does not establish a calibrated probability. Preserve the sealed 66.9% result and test the revised process on a new hidden market target rather than tuning this case to 98.5%.

About this report

This explanation was written at report generation from the recorded research. It does not change the sealed predictions. The complete evidence and calculation records are in the technical appendix.

A Technical appendix

Complete records follow. These do not change the assessment above.

Summary

Workbench workbench_1ae01f8856bf4dc5926f: latest estimate 66.9%; revealed.

Question and resolution

Created at: 2026-09-16T16:40:06.267936Z

Specification:

Domain: economics

Event deadline: 2026-10-14T12:30:00Z

Event group: KXCPI-26SEP

Id: cpi_sep2026_above_0_2

Kind: real

No: The qualifying expiration value is 0.2% or lower.

Profile: general

Resolution source: BLS September CPI release, scheduled October14 at08:30EDT; Kalshi KXCPI-26SEP-T0.2 rules and CPI contract PDF control rounding, delays, expiration and fallback. Market close08:25EDT differs from release time.

Resolve after: 2026-10-14T12:31:00Z

Text: Will the published seasonally adjusted monthly CPI-U increase for September 2026 exceed 0.2%?

Void: Nonbinary discretionary settlement or unresolved ambiguity about a contingency is outside this binary comparison.

Yes: The Kalshi expiration value for September 2026 monthly seasonally adjusted all-items CPI-U, published to one decimal place by BLS, is strictly greater than 0.2%; ordinarily a reported 0.3% or higher. The published contract fallback applies if no underlying is released.

Version: 1

Workbench overview

Case id: workbench_1ae01f8856bf4dc5926f

State: revealed

Question version: 1

Method: Inflation momentum plus a declared gasoline contribution adjustment; hidden market comparison

Mode: prospective

Latest estimate: 66.9%

Sealed at: 2026-09-16T16:47:48.251320Z

Market status: Revealed

Opening target captured at: 2026-09-16T16:41:57.200854Z

Frozen contract and case selection

Question:

Created at: 2026-09-16T16:40:06.267936Z

Specification:

Domain: economics

Event deadline: 2026-10-14T12:30:00Z

Event group: KXCPI-26SEP

Id: cpi_sep2026_above_0_2

Kind: real

No: The qualifying expiration value is 0.2% or lower.

Profile: general

Resolution source: BLS September CPI release, scheduled October14 at08:30EDT; Kalshi KXCPI-26SEP-T0.2 rules and CPI contract PDF control rounding, delays, expiration and fallback. Market close08:25EDT differs from release time.

Resolve after: 2026-10-14T12:31:00Z

Text: Will the published seasonally adjusted monthly CPI-U increase for September 2026 exceed 0.2%?

Void: Nonbinary discretionary settlement or unresolved ambiguity about a contingency is outside this binary comparison.

Yes: The Kalshi expiration value for September 2026 monthly seasonally adjusted all-items CPI-U, published to one decimal place by BLS, is strictly greater than 0.2%; ordinarily a reported 0.3% or higher. The published contract fallback applies if no underlying is released.

Version: 1

Contract:

Event group: KXCPI-26SEP

Market id: KXCPI-26SEP-T0.2

Resolution source: See frozen Kalshi rules and series contract.

Rules: If the Consumer Price Index (CPI) increases by more than 0.2% (single-decimal) in September 2026, then the market resolves to Yes.

The market will always close at 8:25 AM ET on the scheduled day of the data release (October 14, 2026). Please note: the Expiration Value is the single-decimal value published at the Source Agency. In the case of a delay in data caused by a federal government shutdown impacting the

reliability of the Source Agency, the market's latest Expiration Date will be extended to the sooner of the release of the Underlying or six months after the end of the government shutdown.

Scheduled close: 2026-10-14T12:25:00Z

Title: Will CPI rise more than 0.2% in September 2026?

Venue: kalshi

Yes description: Above 0.2%

Selection:

Contract match rationale: Monthly all-items CPI-U seasonally adjusted, signed one-decimal change; strictly above 0.2. Generic terms and market-specific shutdown/close rules retained. Calculation assumes ordinary release; fallback is not independently modeled.

Eligibility rationale: September is incomplete and its BLS CPI release is scheduled for October 14. Select the 0.2% threshold without looking at odds. Require at most 5-cent spread and at least 10 contracts each side.

Market id: KXCPI-26SEP-T0.2

Max spread: 0.05

Method: Inflation momentum plus a declared gasoline contribution adjustment; hidden market comparison

Min contracts each side: 10

Mode: prospective

Question:

Question id: cpi_sep2026_above_0_2

Version: 1

Venue: kalshi

Prediction history

Checkpoints are recorded judgments, not necessarily issued forecasts. All timestamps retain their recorded time zones.

Stage	Initial
Probability	36.9%
Recorded at	2026-09-16T16:42:39.950330Z
Gap to opening market (pp)	-61.56
Squared market error	0.378912
Step improvement	Not applicable

Stage	Research 1: Does current inflation momentum plus observable gasoline pricing put September above the 0.25pp unrounded cutoff, and how uncertain is that estimate?
Probability	18.8%

Recorded at	2026-09-16T16:47:47.928840Z
Gap to opening market (pp)	-79.72
Squared market error	0.635562
Step improvement	-0.256650

Stage	Research 2: Complete the gasoline contribution adjustment specified in the original plan.
Probability	66.9%
Recorded at	2026-09-16T16:47:48.144970Z
Gap to opening market (pp)	-31.58
Squared market error	0.099747
Step improvement	+0.535815

Market comparison

Frozen opening midpoint: 98.5%; bid-ask 98.0%–99.0%. Captured 2026-09-16T16:41:57.200854Z.

Later midpoint: 98.5%; bid-ask 98.0%–99.0%. Captured 2026-09-16T22:28:05.744687Z.

Squared market error = (estimate – opening midpoint)². Positive step improvement means closer agreement.

Market agreement is not an outcome score. The later quote is separate from the frozen opening target.

Target:

Ask: 0.99

Ask contracts within 2c: 2954.5

Ask size: 2954.5

Bid: 0.98

Bid contracts within 2c: 5140.01

Bid size: 595.01

Midpoint: 0.985

Spread: 0.01

Target captured at: 2026-09-16T16:41:57.200854Z

Target method: Opening order-book midpoint, frozen before initial assessment.

Net squared error improvement: 0.279165569248771

Followup quote:

Ask: 0.99

Ask contracts within 2c: 2500.0

Ask size: 2500.0

Bid: 0.98

Bid contracts within 2c: 52884.22

Bid size: 455.16

Midpoint: 0.985

Spread: 0.01

Followup captured at: 2026-09-16T22:28:05.744687Z

Market movement pp: 0.0

Followup error: Not recorded

Contract changed: No

Eligible for blind comparison: Yes

Qualification reasons:

None recorded.

Limitations:

Item 1.

Market agreement is not an outcome score or proof of forecast accuracy.

Item 2.

Step improvements are descriptive; sequential research does not identify causal effects.

Item 3.

Research may include information arriving after the opening target. Inspect market movement and timestamps.

Item 4.

No exposure reported is an agent declaration, not independently verified isolation.

Item 5.

Bid-ask spread is not a confidence interval. Missing follow-up data does not replace the opening target.

Research journal 1: Initial

Recorded 2026-09-16T16:42:39.950330Z · pre_reveal

Entry workbench_entry_9b312ecf943747608ac4

Assumptions:

Item 1.

Ordinary BLS release and usual nearest-tenth rounding

Item 2.

Mean0.20pp, SD0.15pp subjective

Item 3.

No separate probability assigned to shutdown/no-data fallback

Evidence refs:

None recorded.

Probability: 36.9%

Rationale: A deliberately simple starting distribution for unrounded monthly headline CPI: normal mean 0.20 percentage points and SD 0.15 percentage points. Both parameters are subjective macro assumptions, not fitted data. Published single-decimal change must exceed 0.2, approximated by unrounded change ≥ 0.25 . Broad macro context from the prior Fed example was already present; this is not a fresh-context prior.

Research status: in_progress

Uncertainties:

Item 1.

Recent headline and core momentum

Item 2.

Gasoline contribution and seasonal adjustment

Item 3.

Forecast error around a short-run momentum model

Research journal 2: Plan

Recorded 2026-09-16T16:42:40.060260Z · pre_reveal

Entry workbench_entry_a0b06e56ffc1457688f0

Higher if: Recent headline momentum and seasonally adjusted gasoline price movement lift the forecast above 0.25pp.

Lower if: Underlying momentum is softer or gasoline contributes less than in the preceding three months.

Search plan: Fix this primary model before data: compute unrounded monthly rates from headline CPI-U seasonally adjusted indices. Baseline center is mean June, July, August 2026 monthly rates. Estimate bias and residual SD for the same three-prior-month mean rule on all 48 target months September 2022–August 2026, without outcome filtering. Primary spread = $s \cdot \sqrt{1 + 1/48}$; normal residual distribution. First checkpoint is this calibrated momentum model. Then adjust only gasoline: project September US all-grades retail gasoline average using weekly EIA/FRED GASALLW observations available by September 16, carrying each Monday observation forward until the next and carrying the last observed price through September 30. Compute August and projected September calendar-day averages using the same step rule. Translate raw gasoline change to seasonal-adjusted change by adding median September correction (CPI gasoline SA monthly change minus gasoline NSA change) for all five Septembers 2021–2025. Replace the average June–August SA gasoline contribution in the momentum model with this projected contribution, using August 2026 BLS gasoline relative importance divided by 100. Adjustment = $w_{\text{gas}} \cdot (\text{projected gasoline SA change} - \text{mean June–August gasoline SA change})$. Other components retain their implicit three-month trend. Keep the historical momentum-model spread; do not claim new evidence auto-

matically tightens it. Record a second checkpoint using that fixed calculation. Check BLS release and component table, seasonality definitions, CPI index histories, and EIA weekly retail gasoline definitions. Report empirical residual estimate, a chronological36/12 holdout diagnostic, gasoline remaining-month price sensitivity +/-5%, seasonal correction across all five individual years, and center shifts +/-0.05pp. No market odds or third-party event probabilities. Do not switch estimators to improve market agreement.

Stopping rule: Complete these specified datasets and two checkpoints. If an input cannot be obtained, retain the last supported model and disclose the missing calculation. No guessing missing weight or seasonality. Minimum36 valid residual targets. Primary historical data may be revised as of today: disclose that they do not recreate original as-released CPI vintages. Stop and seal before revealing price.

Uncertainty: Does current inflation momentum plus observable gasoline pricing put September above the0.25pp unrounded cutoff, and how uncertain is that estimate?

Why it matters: The release threshold is narrow; energy can move headline CPI by several hundredths of a percentage point. A point forecast alone cannot answer the contract.

Research journal 3: Checkpoint

Recorded 2026-09-16T16:47:47.928840Z · pre_reveal

Entry workbench_entry_0679ebebe89a48ab8c88 · Research plan workbench_entry_a0b06e56ffc1457688f0

Changed assumptions:

Item 1.

Replace subjective0.20pp center and0.15pp SD with the predeclared empirical momentum model.

Item 2.

Use actual valid sample size43 in the1+1/n mean-uncertainty term.

Evidence refs:

Packet id	Record id
pkt_320e7ec936174cc70c759ac3	momentum
pkt_320e7ec936174cc70c759ac3	holdout

Findings: Missing October2025 headline data invalidate five targets; no imputation. Recent holdout RMSE is much worse than earlier fitted spread.

Limitations:

Item 1.

Historical indices are revised; October2025 missing data remove five residual targets.

Item 2.

Gaussian stable residual distribution and three-month momentum are declared modeling assumptions.

Item 3.

Gasoline is a linear contribution approximation; EIA prices, sampling and weights differ from BLS gasoline CPI.

Item 4.

September remainder is projected flat at last observed price; no oil-price, inventory or shock model.

Item 5.

Seasonal correction is five-year September median, not the exact published 2026 adjustment factor.

Item 6.

August release reports July relative importance; primary uses its 3.770% lagged weight, with updated-weight sensitivity.

Item 7.

Residual spread is from the momentum model, not a validated gasoline-adjusted model.

Item 8.

Core, shelter, food and other energy keep implicit three-month trend; no separate forward-looking models.

Item 9.

No model for exceptional delayed-release or no-data fallback; ordinary release assumed.

Item 10.

Same assistant context and procedural price separation, not verified independent blinding.

Item 11.

Source retrieval timestamps conservatively use bundle capture time.

Item 12.

The CPI summary/table/FAQ were read through the web tool after direct HTTP 403 failures; their relevant facts were transcribed, not saved as complete raw HTML.

Item 13.

Gasoline evidence was collected in the same research pass before the momentum checkpoint; stages isolate calculations, not independent experiments.

Item 14.

Costs and model usage are unmetered.

Probability: 18.8%

Rationale: First stage of the declared model: use recent three-month headline momentum and 43 available historical prediction residuals. Mean 0.01472pp and SD 0.26552pp give 18.78% above the effective 0.25pp threshold. Large June decline depresses the raw trend. This intermediate calculation omits already-collected gasoline evidence; it is not an independent research run.

Remaining uncertainties:

Item 1.

Gasoline adjustment still to incorporate

Item 2.

Historical regime and revised data

Item 3.

Seven-observation holdout is small and shows poor coverage

Sources checked:

Item 1.

<https://fred.stlouisfed.org/graph/fredgraph.csv?id=CPIAUCSL,CUSR0000SETB01,CUUR0000SETB01&cosd=2021-08-01&coed=2026-08-01>

Item 2.

<https://www.bls.gov/news.release/cpi.nr0.htm>

Item 3.

<https://www.bls.gov/cpi/seasonal-adjustment/questions-and-answers.htm>

Research journal 4: Plan

Recorded 2026-09-16T16:47:48.035648Z · pre_reveal

Entry workbench_entry_615f5ca864c348a5a918

Higher if: Current September gasoline growth exceeds the prior three-month gasoline trend.

Lower if: Remaining-month prices decline or the seasonal contribution is smaller.

Search plan: Use the inputs already collected under the initial pre-data plan and its same formula; this is a continuation, not a claim these inputs are still unseen. Technical clarification: the August BLS release labels 3.770% as July relative importance, so use that published lagged weight and show a rolled-forward August-weight sensitivity. Retain original residual spread for primary model. Add a diagnostic using recent holdout RMSE as an alternative width; do not claim it is calibrated or select it to match the hidden quote.

Stopping rule: Record the gasoline-adjusted model and declared sensitivities, flag the weak uncertainty calibration, then seal.

Uncertainty: Complete the gasoline contribution adjustment specified in the original plan.

Why it matters: The three-month trend averages a sharp June gasoline decline; current gasoline is rising.

Research journal 5: Checkpoint

Recorded 2026-09-16T16:47:48.144970Z · pre_reveal

Entry workbench_entry_5ce6787d6f6741b1bac5 · Research plan workbench_entry_615f5ca864c348a5a918

Changed assumptions:

Item 1.

Replace only the gasoline contribution with observed partial-month prices plus declared flat remainder.

Item 2.

Five-year September median correction approximates gasoline seasonality.

Item 3.

Lagged published weight 3.770%, rather than mistakenly labeling it an August observation.

Evidence refs:

Packet id	Record id
pkt_320e7ec936174cc70c759ac3	fresh_gas
pkt_320e7ec936174cc70c759ac3	gas_adjustment
pkt_320e7ec936174cc70c759ac3	final
pkt_320e7ec936174cc70c759ac3	holdout

Findings: Primary EIA data extend one week beyond FRED. The implied 4.22% raw gasoline rise plus September seasonal adjustment reverses the backward-looking gasoline drag. Lagged 3.770% weight is transparent; rolled-forward weight sensitivity gives 67.97%.

Limitations:

Item 1.

Historical indices are revised; October 2025 missing data remove five residual targets.

Item 2.

Gaussian stable residual distribution and three-month momentum are declared modeling assumptions.

Item 3.

Gasoline is a linear contribution approximation; EIA prices, sampling and weights differ from BLS gasoline CPI.

Item 4.

September remainder is projected flat at last observed price; no oil-price, inventory or shock model.

Item 5.

Seasonal correction is five-year September median, not the exact published 2026 adjustment factor.

Item 6.

August release reports July relative importance; primary uses its 3.770% lagged weight, with updated-weight sensitivity.

Item 7.

Residual spread is from the momentum model, not a validated gasoline-adjusted model.

Item 8.

Core, shelter, food and other energy keep implicit three-month trend; no separate forward-looking models.

Item 9.

No model for exceptional delayed-release or no-data fallback; ordinary release assumed.

Item 10.

Same assistant context and procedural price separation, not verified independent blinding.

Item 11.

Source retrieval timestamps conservatively use bundle capture time.

Item 12.

The CPI summary/table/FAQ were read through the web tool after direct HTTP403 failures; their relevant facts were transcribed, not saved as complete raw HTML.

Item 13.

Gasoline evidence was collected in the same research pass before the momentum checkpoint; stages isolate calculations, not independent experiments.

Item 14.

Costs and model usage are unmetered.

Probability: 66.9%

Rationale: Apply the originally declared gasoline replacement. Add 0.351478pp to the momentum center for final mean 0.366199pp; keep SD 0.265519pp rather than asserting new evidence justifies tighter uncertainty. Normal tail above 0.25pp gives 66.9173%. This model is provisional: poor holdout coverage challenges stationary uncertainty. Using its 0.50164pp RMSE as an alternative width gives 59.16%, a diagnostic not a fitted replacement.

Remaining uncertainties:

Item 1.

Remaining-month prices and shocks

Item 2.

EIA-to-BLS gasoline mapping and exact seasonal factor

Item 3.

Other CPI components retain backward-looking trends

Item 4.

Historical residual spread may not describe the current regime

Sources checked:

Item 1.

<https://fred.stlouisfed.org/graph/fredgraph.csv?id=CPIAUCSL,CUSR0000SETB01,CUUR0000SETB01&cosd=2021-08-01&coed=2026-08-01>

Item 2.

<https://www.bls.gov/news.release/cpi.nr0.htm>

Item 3.

<https://www.bls.gov/news.release/cpi.t01.htm>

Item 4.

https://www.eia.gov/dnav/pet/hist_xls/EMM_EPM0_PTE_NUS_DPGw.xls

Item 5.

<https://fred.stlouisfed.org/graph/fredgraph.csv?id=GASALLW&cosd=2026-07-01&coed=2026-09-16>

Item 6.

<https://www.bls.gov/cpi/seasonal-adjustment/questions-and-answers.htm>

Item 7.

https://www.eia.gov/petroleum/gasdiesel/gas__proc-methods.php

Item 8.

https://assets.kalshi.com/contract_terms/CPI.pdf

Research journal 6: Finish

Recorded 2026-09-16T16:47:48.251320Z · pre_reveal

Entry workbench_entry_cbde94ad40694d69b776

Exposure notes: No CPI market price or outside event probability observed before seal. Same existing assistant context, including prior Fed-study macro context. Evaluator file store not inspected; no enforced isolation.

Market exposure: none

Outcome status: unresolved

Stopping reason: Completed the fixed momentum model, primary-source gasoline update, seasonal proxy, lagged-weight clarification and sensitivity diagnostics. Report a provisional estimate with weak uncertainty validation; further method changes belong on a fresh target.

Research journal 7: Reflection

Recorded 2026-09-16T22:29:38.890291Z · post_reveal

Entry workbench_entry_1bca3421c6164ef2a880

Next method change: On a fresh hidden target, use a component model with explicit food, shelter/core and energy forecasts. Calibrate the complete model on dated historical partial-month inputs and first-release outcomes, not revised data or baseline-model errors. Add a source-freshness requirement comparing mirrors with the primary release. Use current published seasonal factors if available, and record reasons for exceptional-release assumptions. Do not choose a smaller SD merely to fit this revealed 98.5% quote. Under the user's market-target criterion, this run is a substantial miss despite the useful gasoline update.

What did not: The sealed forecast remains 31.6 points below the opening midpoint. Three-month headline momentum is distorted by large changing gasoline contributions; its historical residual variance is not the conditional uncertainty of the gasoline-adjusted model. Recent holdout coverage was poor. We did not validate the full adjusted model, model other CPI components forward, use exact 2026 seasonal factors, or forecast the remaining gasoline-price path. These are concrete shortcomings, not evidence that Kalshi is wrong.

What helped: The primary EIA workbook supplied a week missing from FRED, and explicit gasoline seasonality/contribution arithmetic changed the estimate from 18.8% momentum-only to 66.9%.

This moved markedly closer to the opening 98.5% midpoint. Quote depth passed the predeclared screen without relaxing it.

Model and calculation

No structured model artifact was attached to this case. The recorded rationales and assumptions above are the available model description. Supplemental files, if supplied, are identified separately below.

Limitations and research cost

Case limitations:

Item 1.

Prices are separated from this project, not protected by an operating-system sandbox.

Item 2.

Selection checks spread and top-of-book size only; neither establishes market accuracy.

Item 3.

The agent must verify the event is still unknown; an open market alone cannot establish this.

Reported usage:

None recorded.

Usage note: Usage is agent-reported. Missing usage is unmetered, not zero; partial records are not a total.

Supplemental model or research file: calculation.json

Supplied at report generation. This is not automatically part of the sealed research record; its hash can be compared with a previously recorded commitment. No code in this file was executed.

Name: calculation.json

Sha256: f6eee597d3fd622fe665ef6378b253d61c113493f8f60002774cd5318ddfb2dd

Provenance: report_time_attachment

Content:

Units: Monthly inflation percentage points, not fractions. Probability is a fraction.

Initial probability: 0.36944134018176367

N: 43

Pairs:

Month	Actual change pp	Three month forecast pp	Error pp
2022-09-01	0.42426727482827165	0.434525283831666	- 0.010258009003394353

Month	Actual change pp	Three month forecast pp	Error pp
2022-10-01	0.5594754832984217	0.15724704256090036	0.4022284407375213
2022-11-01	0.2614032556282231	0.34871134654108626	-
			0.08730809091286318
2022-12-01	0.01539563433359259	0.4150486712516388	-0.3996530369180462
2023-01-01	0.5314022594635093	0.27875812442007913	0.25264413504343014
2023-02-01	0.3428533386591992	0.269400383141775	0.0734529555174242
2023-03-01	0.12307181953890023	0.29655041081876704	-0.1734785912798668
2023-04-01	0.3392739405144063	0.33244247255386955	0.006831467960536741
2023-05-01	0.1614687381333635	0.26839969957083526	-
			0.10693096143747177
2023-06-01	0.224175331482801	0.20793816606222335	0.016237165420577654
2023-07-01	0.1957146710348745	0.24163933671019025	-
			0.04592466567531575
2023-08-01	0.4835707415079771	0.19378624688367965	0.28978449462429745
2023-09-01	0.3900915440960384	0.3011535813418842	0.08893796275415422
2023-10-01	0.13668493471667986	0.3564589855462967	-
			0.21977405082961682
2023-11-01	0.14689823722116024	0.33678240677356513	-0.1898841695524049
2023-12-01	0.1924399963653789	0.22455823867795952	-
			0.03211824231258062
2024-01-01	0.30996854969052023	0.15867438943440634	0.1512941602561139
2024-02-01	0.40975401843086345	0.21643559442568647	0.193318424005177
2024-03-01	0.4431338373525273	0.3040541881622542	0.1390796491902731
2024-04-01	0.21706766556210955	0.387618801824637	-
			0.17055113626252744
2024-05-01	0.04855873210594108	0.3566518404485001	-
			0.30809310834255904
2024-06-01	-	0.23625341167352598	-
	0.041829647960411886		0.27808305963393787
2024-07-01	0.16770805381991494	0.07459891656921291	0.09310913725070202
2024-08-01	0.1572221743858515	0.05814571265514804	0.09907646173070347
2024-09-01	0.2133336729690294	0.09436686008178485	0.11896681288724453
2024-10-01	0.2856398459641696	0.17942130039159862	0.10621854557257096
2024-11-01	0.28419261732848256	0.21873189777301683	0.06546071955546573
2024-12-01	0.3399383308901438	0.26105537875389384	0.07888295213624996
2025-01-01	0.42726162139017365	0.3032569313942653	0.12400468999590836
2025-02-01	0.22510589068882592	0.3504641898696	-
			0.12535829918077407
2025-03-01	0.03315826188146076	0.33076861432304777	-0.297610352441587
2025-04-01	0.16167112278562268	0.22850859132015344	-
			0.06683746853453076
2025-05-01	0.09928130327003792	0.13997842511863645	-
			0.04069712184859853
2025-06-01	0.2541949971929469	0.09803689597904046	0.15615810121390644
2025-07-01	0.22835098853577485	0.17171580774953585	0.056635180786239
2025-08-01	0.3482644202266627	0.19394242966625322	0.1543219905604095

Month	Actual change pp	Three month forecast pp	Error pp
2025-09-01	0.2950901819104068	0.27693680198512816	0.018153379925278657
2026-03-01	0.8651438343614481	0.24521138428212166	0.6199324500793265
2026-04-01	0.6400377846336402	0.43432985283440306	0.2057079317992372
2026-05-01	0.47291422864139676	0.5907282310682641	-
			0.11781400242686735
2026-06-01	-	0.6593652825454951	-1.08184693558356
	0.42248165303806484		
2026-07-01	0.07366914435544825	0.2301567867456574	-
			0.15648764239020915
2026-08-01	0.3960181843858157	0.04136723998626005	0.35465094439955563

Excluded:

Month	Reason
2025-10-01	Missing target or one of three lagged monthly changes; no interpolation.
2025-11-01	Missing target or one of three lagged monthly changes; no interpolation.
2025-12-01	Missing target or one of three lagged monthly changes; no interpolation.
2026-01-01	Missing target or one of three lagged monthly changes; no interpolation.
2026-02-01	Missing target or one of three lagged monthly changes; no interpolation.

Recent headline changes pp:

2026-06: -0.42248165303806484

2026-07: 0.07366914435544825

2026-08: 0.3960181843858157

Raw momentum center pp: 0.015735225234399703

Mean residual pp: -0.0010144370038234426

Sample residual sd pp: 0.26248394571848377

Predictive sd pp: 0.2655185430145959

Momentum center pp: 0.01472078823057626

Momentum probability: 0.1877785433996466

Holdout:

Train n: 36

Test n: 7

Train bias pp: 0.0031689745286897367

Train sd pp: 0.17983254839871332

Raw rmse pp: 0.5016401879054787

Corrected rmse pp: 0.5017924959255947

Nominal90 coverage: 0.5714285714285714

Halfwidth pp: 0.2998783876611614

Qualification: Fixed first36/last12 calendar months; missing data reduce valid holdout to seven.
Revised-vintage diagnostic, not original-release validation.

Gasoline:

Latest observation: 2026-09-14

Latest price: 4.455

August daily step average: 4.191193548387097

September projected daily step average: 4.3680666666666665

Raw projected change pct: 4.22011334569925

Seasonal corrections:

Year	Sa change pct	Nsa change pct	Correction pp
2021	2.394040566415989	0.32078083100257615	2.073259735413413
2022	-3.7216736795961047	-5.644117925489045	1.9224442458929403
2023	2.7899203277098383	0.5679879161609502	2.221932411548888
2024	-2.8270912930686842	-5.06482727606401	2.237735982995326
2025	3.5936542621123335	1.1464169600294838	2.4472373020828497

Median seasonal correction pp: 2.221932411548888

Projected sa change pct: 6.442045757248138

Recent three month sa change pct: -2.8809794370296

Weight fraction: 0.0377

Weight reference month: 2026-07

Headline adjustment pp: 0.3514780498242707

Final center pp: 0.366198838054847

Final probability: 0.6691726825804136

Empirical residual probability: 0.7209302325581395

Remaining month gas sensitivity:

Future price shift	Center pp	Probability
-0.05	0.2726952822119506	0.5340582419535114
0	0.366198838054847	0.6691726825804136
0.05	0.4597023938977433	0.7851731442463052

Seasonal sensitivity:

Year	Center pp	Probability
2021	0.3605938781645396	0.6614853626020221
2022	0.35490813420961775	0.6536178984474103
2023	0.366198838054847	0.6691726825804136
2024	0.3667946326983777	0.6699857193865957
2025	0.3746928324279773	0.6806867526595564

Center sensitivity:

Shift pp	Probability
-0.05	0.5984430120545602
0	0.6691726825804136
0.05	0.7343230507843395

Rolled weight sensitivity:

Weight fraction: 0.038532934906495134

Probability: 0.6797058036818716

Label: Approximate August weight rolled forward with NSA relative changes; not selected primary.

Limitations:

Item 1.

Historical indices are revised; October2025 missing data remove five residual targets.

Item 2.

Gaussian stable residual distribution and three-month momentum are declared modeling assumptions.

Item 3.

Gasoline is a linear contribution approximation; EIA prices, sampling and weights differ from BLS gasoline CPI.

Item 4.

September remainder is projected flat at last observed price; no oil-price, inventory or shock model.

Item 5.

Seasonal correction is five-year September median, not the exact published2026 adjustment factor.

Item 6.

August release reports July relative importance; primary uses its3.770% lagged weight, with updated-weight sensitivity.

Item 7.

Residual spread is from the momentum model, not a validated gasoline-adjusted model.

Item 8.

Core, shelter, food and other energy keep implicit three-month trend; no separate forward-looking models.

Item 9.

No model for exceptional delayed-release or no-data fallback; ordinary release assumed.

Item 10.

Same assistant context and procedural price separation, not verified independent blinding.

Raw sha256:

Bls weight.json: 2dd141c4a1d4c470a33721ea9cf2e7d0d4388391cb4d176d8d3b4a28773cbfdb

Contract.pdf: 2317b1d8e823082b409f6ff3415fb135804d9682681f9f92f640b3681b29a872

Contract.txt: 72aa0a88d67087b387a4294f17eee41eaaa1aef6a01caa9f7490286cfb047bdf

Cpi indices.csv: beba624d05a9b5106081951632efe5b41047fef956ac1b0ad59246f68f9cb0dc

Eia gasoline weekly.xls: a6f774ea70a2b0f32db48e4255ba5ff7fe7d99c106320d22683ab5a9d64e7aac

Eia method.html: 6bb5835afeb9681731b884f73ea6c96b54715eebaadb319addcaeb73ab341f4

Gasoline primary.csv: b8ae43b752def92b626764ef54112762625b6e01656edc0f7fa8acf28ddd2b5b

Gasoline weekly.csv: 52f2068c19c092b36af9e3a50f60ca13da9d02b86a735bc1c84cdd68e40abe57

Series.json: f5c410bc20a280d5fc14e33d1b028777a3a88aa6a955938eeee03cc481866e60

Finding: pkt_320e7ec936174cc70c759ac3:momentum

Sources: [1], [2], [3]

Claim: Fixed three-month momentum center 0.015735pp; mean residual -0.001014pp; 43 valid residual targets out of 48, excluding October 2025–February 2026 due to the missing CPI index. SD 0.265519pp gives 18.777854% above 0.25pp.

Claim type: inference

Entity ids:

None recorded.

Id: momentum

Observed at: 2026-09-16T16:47:47.744196+00:00

Provenance:

Adapter: vorhersage.research_bundle.v1

Value: Not recorded

Finding: pkt_320e7ec936174cc70c759ac3:holdout

Sources: [1], [3]

Claim: Fixed 36-month training and 12-calendar-month holdout leave 36 and 7 valid observations. Held-out RMSE 0.5016pp; bias correction does not improve it. Nominal 90% interval covers 4/7 = 57.1%. This raises a calibration/regime warning, not a validated accuracy claim.

Claim type: inference

Entity ids:

None recorded.

Id: holdout

Observed at: 2026-09-16T16:47:47.744196+00:00

Provenance:

Adapter: vorhersage.research_bundle.v1

Value: Not recorded

Finding: pkt_320e7ec936174cc70c759ac3:fresh_gas

Sources: [4], [5], [6]

Claim: FRED stopped September7; the EIA workbook adds September14 all-grades\$4.455/gal. Carrying the last price through month-end gives a September average\$4.3681 versus August\$4.1912, a4.2201% increase.

Claim type: inference

Entity ids:

None recorded.

Id: fresh_gas

Observed at: 2026-09-16T16:47:47.744196+00:00

Provenance:

Adapter: vorhersage.research_bundle.v1

Value: Not recorded

Finding: pkt_320e7ec936174cc70c759ac3:gas_adjustment

Sources: [1], [7], [4], [3]

Claim: September seasonal correction from all five2021–2025 Septembers has median+2.2219 percentage points. Projected gasoline SA growth6.4420% replaces the prior three-month average-2.8810%; a3.770% lagged weight adds0.351478pp to headline prediction. Linear contribution is approximate.

Claim type: inference

Entity ids:

None recorded.

Id: gas_adjustment

Observed at: 2026-09-16T16:47:47.744196+00:00

Provenance:

Adapter: vorhersage.research_bundle.v1

Value: Not recorded

Finding: pkt_320e7ec936174cc70c759ac3:final

Sources: [1], [4], [7], [8]

Claim: Primary final headline center 0.366199pp and retained momentum-model SD0.265519pp imply66.917268% probability. Remaining-month gasoline prices +/-5% give53.41-78.52%. Calculation SHA256 f6eee597d3fd622fe665ef6378b253d61c113493f8f60002774cd5318ddfb2dd

Claim type: inference

Entity ids:

None recorded.

Id: final

Observed at: 2026-09-16T16:47:47.744196+00:00

Provenance:

Adapter: vorhersage.research_bundle.v1

Value: Not recorded

Source [1]: BLS CPI indices via Federal Reserve Bank of St Louis

Excerpt: Downloaded headline SA and gasoline SA/NSA monthly index levels through August2026. October2025 headline is missing. The calculation filters observations to the predeclared window.

Excerpt kind: paraphrase

Id: indices

Retrieved at: 2026-09-16T16:47:47.744196+00:00

Title: BLS CPI indices via Federal Reserve Bank of St Louis

Url: <https://fred.stlouisfed.org/graph/fredgraph.csv?id=CPIAUCSL,CUSR0000SETB01,CUUR0000SETB01&cosd=2021-08-01&coed=2026-08-01>

Source [2]: BLS August2026 CPI release

Excerpt: Published September11. Headline monthly changes were -0.4%,+0.1%,+0.4% for June-August; August core rose0.3%. Gasoline increased3.9% in August. September release scheduled October14. Seasonal data are revised annually.

Excerpt kind: paraphrase

Id: bls

Published at: 2026-09-11T12:30:00Z

Retrieved at: 2026-09-16T16:47:47.744196+00:00

Title: BLS August2026 CPI release

Url: <https://www.bls.gov/news.release/cpi.nr0.htm>

Source [3]: BLS seasonal adjustment FAQ

Excerpt: Seasonality removes recurring calendar effects. Factors and historical seasonally adjusted series are revised. October2025 was unavailable for many series because of the funding lapse; special methods were used in seasonal estimation.

Excerpt kind: paraphrase

Id: seasonal

Retrieved at: 2026-09-16T16:47:47.744196+00:00

Title: BLS seasonal adjustment FAQ

Url: <https://www.bls.gov/cpi/seasonal-adjustment/questions-and-answers.htm>

Source [4]: EIA weekly US all-grades all-formulations retail gasoline price history

Excerpt: Latest observed Monday September14 price is\$4.455/gal, versus\$4.295 September7 and\$4.207 August31. Archived original workbook and extracted recent rows.

Excerpt kind: paraphrase

Id: eia

Retrieved at: 2026-09-16T16:47:47.744196+00:00

Title: EIA weekly US all-grades all-formulations retail gasoline price history

Url: https://www.eia.gov/dnav/pet/hist_xls/EMM_EPM0_PTE_NUS_DPGw.xls

Source [5]: FRED EIA weekly gasoline mirror

Excerpt: Downloaded series ended September7 at\$4.295. Primary EIA workbook supplies the missing September14 observation; do not treat the mirror as current.

Excerpt kind: paraphrase

Id: fredgas

Retrieved at: 2026-09-16T16:47:47.744196+00:00

Title: FRED EIA weekly gasoline mirror

Url: <https://fred.stlouisfed.org/graph/fredgraph.csv?id=GASALLW&cosd=2026-07-01&coed=2026-09-16>

Source [6]: EIA retail gasoline survey methodology

Excerpt: Weekly retail prices are survey estimates. EIA gasoline measurement and CPI sampling are not identical; calendar-day step averages are our approximation, not an official monthly CPI estimator.

Excerpt kind: paraphrase

Id: eiamethod

Retrieved at: 2026-09-16T16:47:47.744196+00:00

Title: EIA retail gasoline survey methodology

Url: https://www.eia.gov/petroleum/gasdiesel/gas_proc-methods.php

Source [7]: BLS CPI August2026 Table1

Excerpt: August release reports July2026 gasoline relative importance3.770%. NSA gasoline indices July350.846,August359.738; headline July333.918,August334.980. Published weight is lagged, not an August weight.

Excerpt kind: paraphrase

Id: weight

Published at: 2026-09-11T12:30:00Z

Retrieved at: 2026-09-16T16:47:47.744196+00:00

Title: BLS CPI August2026 Table1

Url: <https://www.bls.gov/news.release/cpi.t01.htm>

Source [8]: Kalshi CPI contract terms

Excerpt: Underlying is signed one-month,one-decimal seasonally adjusted CPI-U. Strictly greater than threshold resolves Yes. Post-expiration revisions are excluded. No-data fallback and discretionary review provisions apply.

Excerpt kind: paraphrase

Id: contract

Retrieved at: 2026-09-16T16:47:47.744196+00:00

Title: Kalshi CPI contract terms

Url: https://assets.kalshi.com/contract_terms/CPI.pdf

Report provenance

This report renders stored records without new research, model calls, or probability revisions. Missing work remains missing. The JSON export preserves the complete report input.

Schema version: vorhersage.report.v1

Generated at: 2026-09-16T22:31:28.278519Z

Record sha256: 1e693802426ef1db3df4b38705474eb1ec94370797176a6347a91ef49e45755a

Format: latex

Latex engine: XeLaTeX or LuaLaTeX

Narrative sha256: f149195f938a47d568f4427840c41ef6adc1c7fdfe6632b0b265c560601cd45a